Lecture 17
Binomial, Poisson and Normal Distributions

Review

1 What is the expected value of a random variable?

The long-run average of X: $\mu = \Sigma\, x \cdot P(x)$

2 What is the standard deviation of a random variable?

The typical distance of X from μ: $\sigma = \sqrt{\Sigma\, (x - \mu)^2 \cdot P(x)}$

3 What is a Bernoulli trial?

One trial with two outcomes: success with probability p, failure with probability 1 − p.

4 What is a binomial random variable?

The number of successes in n independent trials, each with the same probability of success p.

Example: free throws

Stephen Curry has made 4,236 of the 4,645 free throws he has attempted in the NBA.

$p = 4{,}236 / 4{,}645 \approx 0.912$; he shoots $n = 10$; X = free throws made

(a) What is P(he makes all 10)?
(b) What is P(he makes at least 2), $P(X \geq 2)$?
(c) How many does he make, on average?

(a) $0.912^{10} \approx 0.40$
(b) $1 - P(X < 2) = 1 - [P(0) + P(1)] = 1 - [0.088^{10} + 10 \times 0.912 \times 0.088^{9}]$
$\approx 1 - 0.000000003 \approx 1$: he is all but certain to make 2
(c) $\mu = np = 10 \times 0.912 \approx 9.1$

Example: a lymphoma cluster?

A doctor in a rural California county finds that 71 of his 893 patients (about 8%) were diagnosed with lymphoma last year.

US rate ≈ 21 per 100,000 people a year: $p = 0.00021$; X = cases among 893 patients

(a) At the national rate, how many cases should he expect, μ?
(b) Write $P(X \geq 71)$ with the complement rule.
(c) Software gives $P(X \geq 71)$. Should he contact the CDC?

(a) $\mu = 893 \times 0.00021 \approx 0.19$ cases
(b) $P(X \geq 71) = 1 - P(X \leq 70) = 1 - [P(0) + P(1) + \ldots + P(70)]$
(c) $\approx 1 \times 10^{-155}$: essentially impossible at the national rate, so yes

Is it binomial?

For each X, check the three conditions: fixed n, the same p, independent trials.

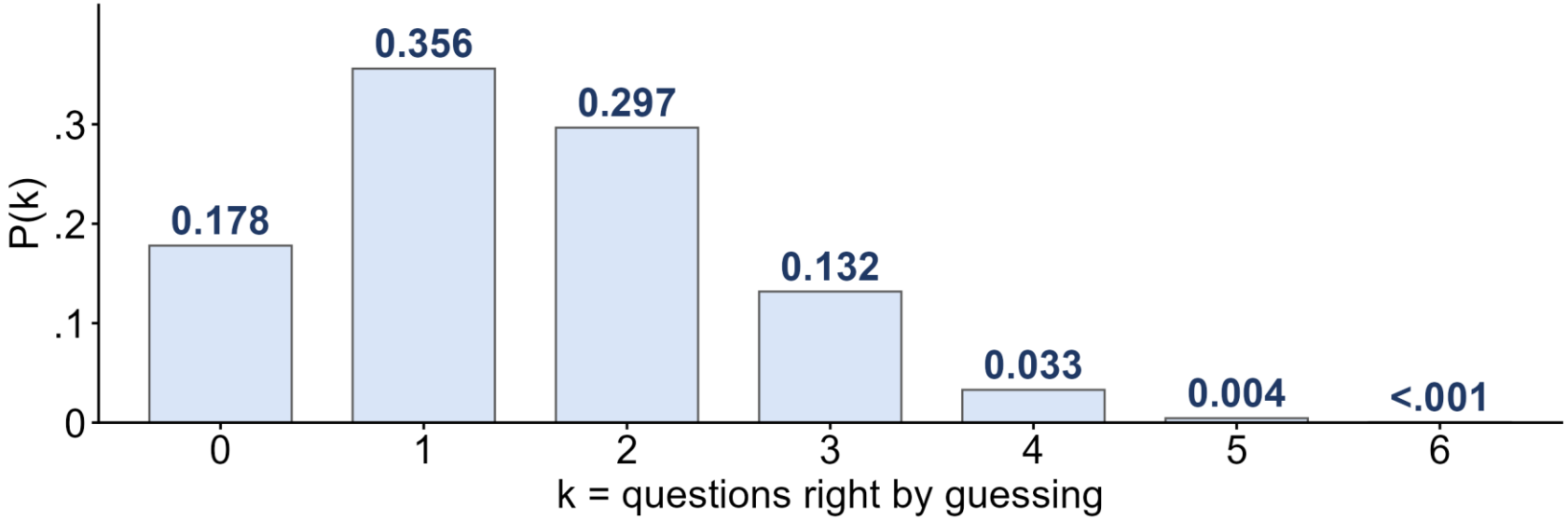
(a) X = the number of heads in 20 flips of a coin
(b) X = the number of aces in a 5-card hand
(c) X = the number of rolls of a die until the first six

(a) Yes: n = 20, p = 0.5, and the flips are independent
(b) No: the cards are dealt without replacement, so the draws are not independent
(c) No: the number of trials n is not fixed

Example: guessing on a quiz

A student guesses on all 6 questions of a quiz; each has 4 options: X ~ Binom(6, 0.25).

Use the bars: (a) P(X ≤ 1)? (b) P(X ≥ 3)? (c) P(1 ≤ X ≤ 3)?



(a) 0.178 + 0.356 = 0.534 (b) 1 − (0.178 + 0.356 + 0.297) = 0.169
(c) 0.356 + 0.297 + 0.132 = 0.785

The Poisson distribution

X ~ Poisson(λ)

X = the number of events in a fixed interval of time or space, when events occur independently at an average rate λ

P(X = k)

$\lambda^k e^{-\lambda} / k!$, k = 0, 1, 2, … (e ≈ 2.718)

mean and variance

$\mu = \lambda$ and $\sigma^2 = \lambda$

Example: World Cup goals

The 2022 FIFA World Cup had 172 goals in 64 matches.

$\lambda = 172 / 64 \approx 2.69$ goals per match; X = goals in one match

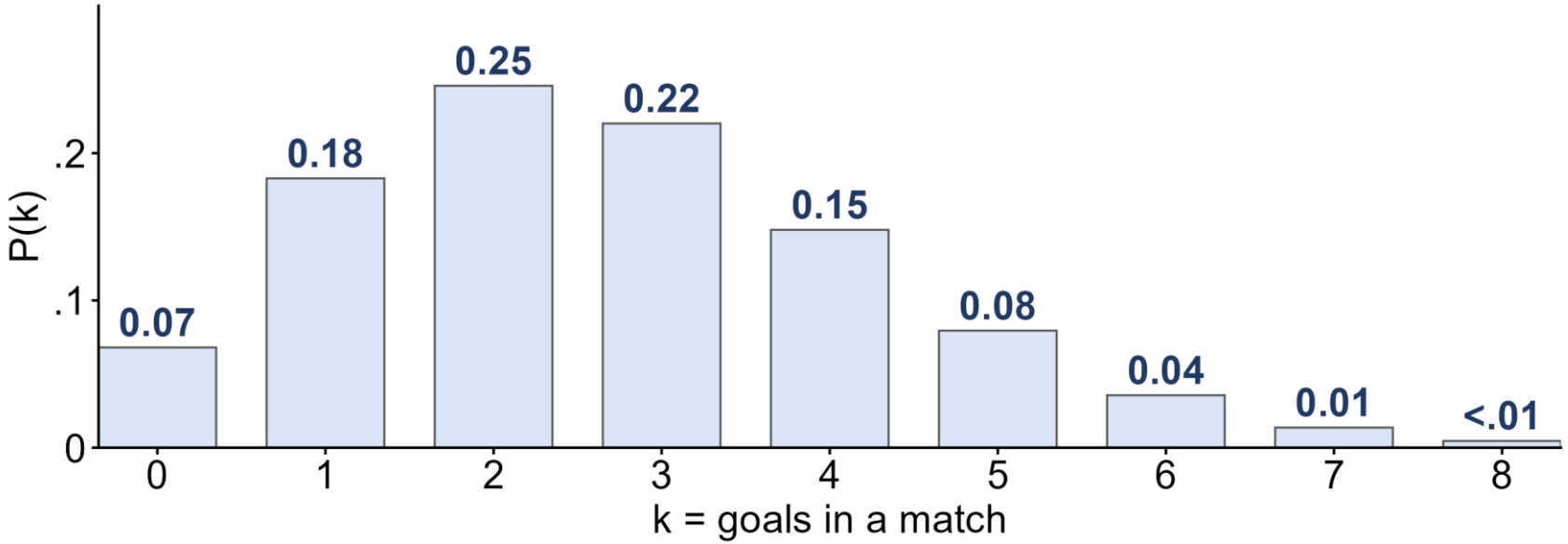
(a) What is $P(X = 0)$?
(b) What is $P(X > 1)$?

(a) $e^{-2.69} \approx 0.068$
(b) $1 - [P(0) + P(1)] = 1 - [0.068 + 2.69 \times 0.068] \approx 0.75$

Goals per match: the Poisson model

X ~ Poisson(λ = 2.69): the probability of each number of goals in a match.

Which number of goals is most likely? Is the distribution symmetric?

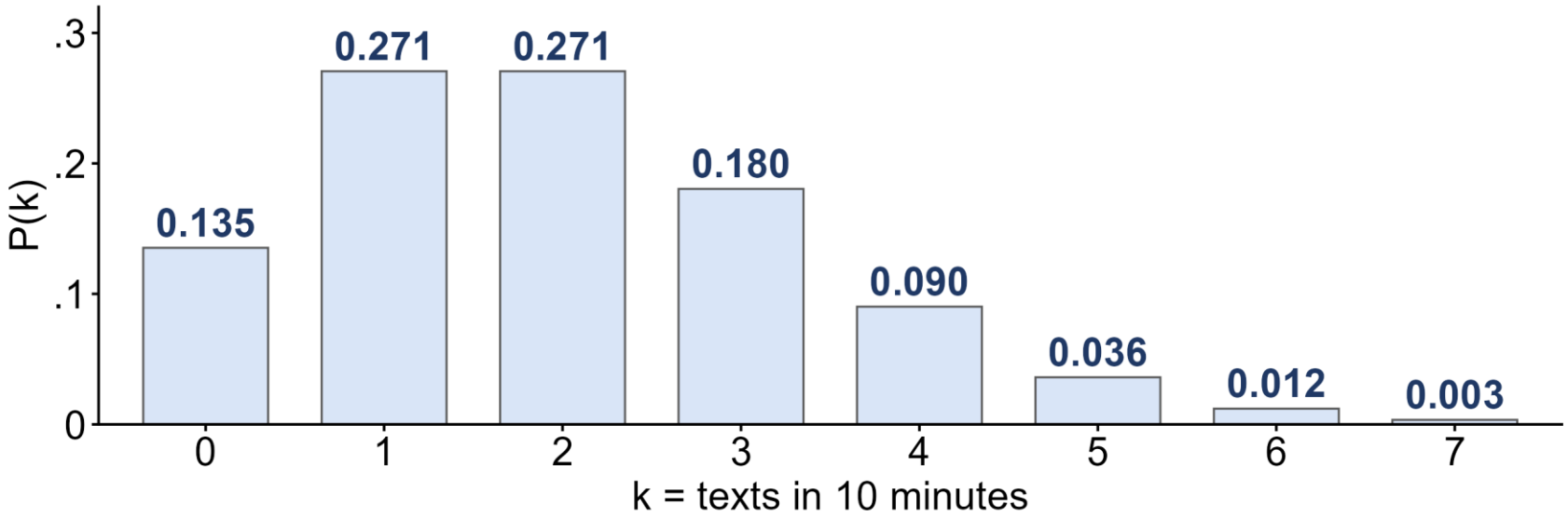


Two goals (0.25), then three (0.22); skewed right: X cannot go below 0 but has no upper limit

Example: texts during class

You get an average of 2 text messages every 10 minutes of class: X ~ Poisson(λ = 2).

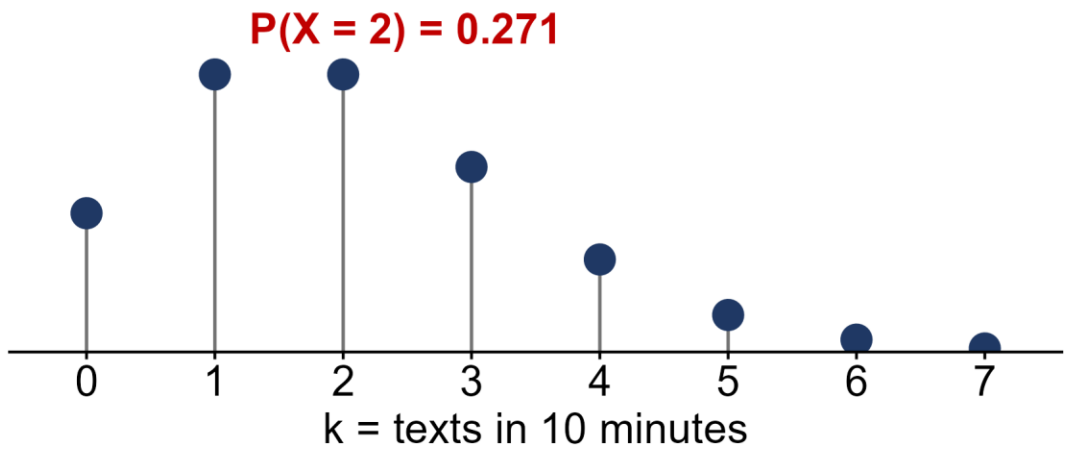
Use the bars: (a) P(X ≤ 1)? (b) P(X ≥ 3)? (c) P(2 ≤ X ≤ 4)?



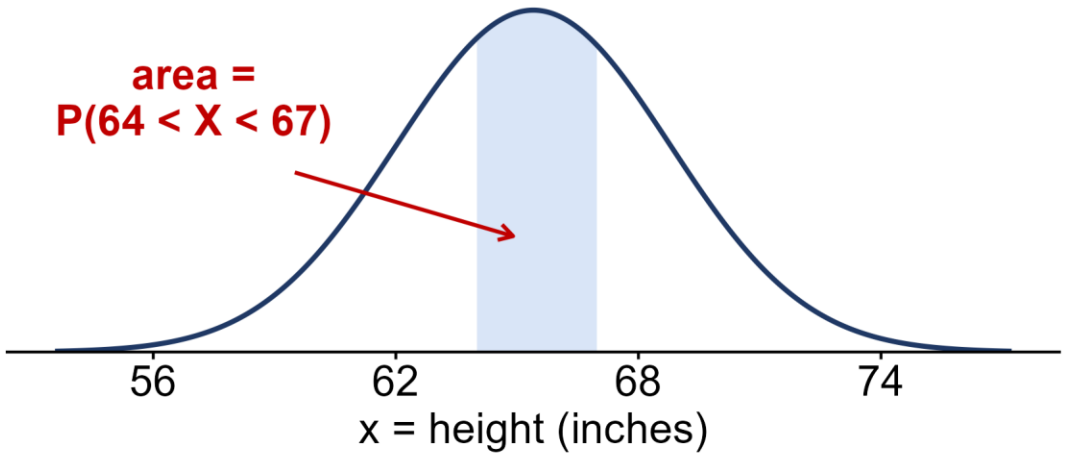
(a) 0.135 + 0.271 = 0.406 (b) 1 − (0.135 + 0.271 + 0.271) = 0.323
(c) 0.271 + 0.180 + 0.090 = 0.541

Probability mass vs. probability density

Discrete: mass



Continuous: density



probability mass (discrete)

each value carries its own packet of probability, P(X = k), and the packets add to 1

probability density (continuous)

how concentrated probability is around a point; probability is the area under the curve

The normal as a continuous distribution

$X \sim N(\mu, \sigma)$

a continuous random variable: it can take any value, and its density curve is the bell curve

probability is area

P(a < X < b) is the area under the curve between a and b; the height is not a probability

$P(X = c) = 0$

a single value has no width, so $P(X < c) = P(X \leq c)$

Cumulative probability

$P(X \leq x)$

the area to the left of x: what the z-table and pnorm give

area to the right

$P(X > x) = 1 - P(X \leq x)$

area between

$P(a < X < b) = P(X \leq b) - P(X \leq a)$

Example: area between

X = the height of a woman in the student heights data, modeled as normal.

X ~ N(65.4, 3.38); Week 6: P(X < 70) = 0.9131

(a) What is P(X > 70)?
(b) What is P(X < 62)?
(c) What is P(62 < X < 70)?

(a) 1 – 0.9131 = 0.0869
(b) z = (62 – 65.4) / 3.38 = –1.01: table gives 0.1562
(c) 0.9131 – 0.1562 = 0.7569 ≈ 0.76

What you should know: distributions

expected value

$\mu = \Sigma\, x\, P(x)$: the long-run average of X

binomial

successes in n independent trials with the same p: $\mu = np$, $\sigma^2 = np(1 - p)$; Bernoulli is $n = 1$

Poisson

events in a fixed interval at an average rate λ: $\mu = \sigma^2 = \lambda$

normal

$X \sim N(\mu, \sigma)$: probability is area under the bell curve; $P(X = c) = 0$; $P(a < X < b) = P(X \le b) - P(X \le a)$